

```

#include <Arduino.h>

#include <WiFi.h>
#include <WiFiMulti.h>
#include <HTTPClient.h>

// PIN configuration
const int fsrPin = 34;           // Analog pin for FSR (ensure
it's an analog pin like GPIO 34, 35, 32, 33)
const int reedPin = 14;         // Digital pin for reed switch

// Thresholds
const int fsrThreshold = 200; // Adjust this depending on
your FSR readings

WiFiMulti wifiMulti;

void setup() {
    Serial.begin(115200);
    while (!Serial) {
        // wait for serial port to connect. Needed for native
        USB port only
        delay(500);
    }

    pinMode(reedPin, INPUT);
    // No need to define analog pin mode for fsrPin

    Serial.println("Connecting WIFI...");
    wifiMulti.addAP("RAMSO", "876543210");
    while ((wifiMulti.run() != WL_CONNECTED)) {
        delay(500);
    }
    Serial.println("WIFI Connected!");
}

void loop() {
    int fsrValue = analogRead(fsrPin);           // Soma thamani ya

```

FSR

```
    int reedState = digitalRead(reedPin);    // Soma reed switch
(Low = closed)

    Serial.print("FSR Value: ");
    Serial.print(fsrValue);
    Serial.print(" | Reed: ");
    Serial.println(reedState == LOW ? "FUNGWA" : "HAIJAFUNGWA");

    // Logic ya kuamua
    if (fsrValue > fsrThreshold) { // Kiti kimekaliwa
        if (reedState == HIGH) { // Mkanda HAUJAFUNGWA
            Serial.println("⚠ Abiria ameketi bila kufunga mkanda!");
        };

        sendData("P01", 0);
        // Hapa unaweza kuongeza vibration motor au LED au
buzzer
    } else {
        Serial.println("✔ Mkanda umefungwa vizuri.");
        sendData("P01", 1);
    }
} else {
    Serial.println("🚫 Hakuna abiria kwenye kiti.");
    sendData("P01", '');
}

delay(1000); // Subiri kidogo kabla ya kusoma tena
}
```

```
void sendData(String seatNo, int seatStatus){
    Serial.println("Sending Data for Seat No: " +
(String)seatNo + ", with Status: " + (String)seatStatus);
    if ((wifiMulti.run() == WL_CONNECTED)) {
        HTTPClient http;
        String url = "http://172.18.8.56/recordSeatStatus.php?
seatNo=" + (String)seatNo + "&seatState=" + (int)seatStatus;
        if (http.begin(url)) {
            // start connection and send HTTP header
```

```
int httpCode = http.GET();
// httpCode will be negative on error
if (httpCode > 0) {
    // HTTP header has been send and Server response
header has been handled
    Serial.println("DATA IS SENT!");
    // file found at server
    if (httpCode == HTTP_CODE_OK) {
        String payload = http.getString();
        Serial.println(payload);
    }
}
http.end();
} else {
    Serial.println("CONNECTION ERROR");
}
}else{
    Serial.println("WIFI DISCONNECTED");
}
delay(5000);
}
```